Can you find which ratio is greater in Fig. 3.2?

Express ratios as a fraction and then find the equivalent fractions, until the denominators are the same, and compare the fractions with common denominators. This is done as follows :

Idly Batter Ratio - (a)	Idly Batter Ratio - (b)
$\frac{1}{4} \times \frac{2}{2} = \frac{2}{8}$	$\frac{1}{3} \times \frac{2}{2} = \frac{2}{6}$
$\frac{1}{4} \times \frac{3}{3} = \frac{3}{12}$	$\frac{1}{3} \times \frac{3}{3} = \frac{3}{9}$
$\frac{1}{4} \times \frac{4}{4} = \frac{4}{16}$	$\frac{1}{3} \times \frac{4}{4} = \frac{4}{12}$

Comparing the equivalent ratios, $\frac{4}{12}$ & $\frac{3}{12}$, we can conclude that 1 : 3 is greater than 1 : 4

Situation 2

Let us consider another situation. For example, if a thread of 5 m is cut at 3 m, then the length of two pieces are 3 m and 2 m and the ratio of the two pieces is 3 : 2. From this we say that, a ratio '$a : b$' is said to have a total of '$a+b$' parts in it.

Example 3.5

Kumaran has ₹600 and wants to divide it between Vimala and Yazhini in the ratio 2 : 3. Who will get more and how much?

Solution

Divide the whole money into 2 + 3 = 5 equal parts then, Vimala gets 2 parts out of 5 parts and Yazhini gets 3 parts out of 5 parts.

Amount Vimala gets = ₹ 600 × $\frac{2}{5}$ = ₹ 240

Amount Yazhini gets = ₹ 600 × $\frac{3}{5}$ = ₹ 360

Vimala received ₹ 240 and Yazhini gets ₹ 360, which is ₹ 120 more than that of Vimala.

W39CV

Exercise 3.2

1. Fill in the blanks for the given equivalent ratios.

   (i) 3 : 5 = 9 : ___ (ii) 4 : 5 = ___ : 10 (iii) 6 : ___ = 1 : 2

2. Complete the table.

(i)

Feet	1	2	3	?
Inch	12	24	?	72

(ii)

Days	28	21	?	63
Weeks	4	3	2	?

6th_Maths_EM_Term1_Chapter 3.indd 54 03-12-2022 12:53:04